# Data Note

AI Exposure and Metropolitan Migration:
Data Sources and Construction

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## Contents



## 1 Data Sources

The metro-year panel is constructed from three sources merged in sequence.

**IPUMS American Community Survey (2014–2024).** Individual-level microdata are drawn from the IPUMS ACS extracts for survey years 2014–2024 (file `Data/Raw/usa_00004.dat`, parsed via the accompanying XML codebook `Data/Raw/usa_00004.xml`). Each observation is a survey respondent. Key variables used are: `MET2013` (metropolitan area, 2013 delineation), `YEAR`, `PERWT` (person weight), `AGE`, `EDUCD` (detailed educational attainment), `MIGRATE1` and `MIGMET131` (migration status and origin metro over the past year), `GQ` (group-quarters status), `SEX`, `LABFORCE`, `EMPSTAT`, and `INCWAGE`.

**FHFA House Price Index.** Annual house price indices at the Core-Based Statistical Area (CBSA) level are taken from the Federal Housing Finance Agency (file `Data/Raw/hpi_at_metro.cs` Quarterly observations are averaged to obtain an annual HPI value per metro-year. Several large metropolitan areas in the FHFA data are reported at the Metropolitan Statistical Area Division (MSAD) level rather than the parent CBSA level. These division codes are mapped to their parent CBSAs using the Census Bureau Metropolitan Area Delineation File, covering 13 large metros (Atlanta, Boston, Chicago, Dallas, Detroit, Los Angeles, Miami, New York, Philadelphia, San Francisco, Seattle, Tampa, and Washington DC).

**Metro AI Exposure Index.** A pre-determined, metro-level AI exposure score is taken from `Data/Raw/metro_ai_exposure.csv`. The index is constructed as a population-weighted average of occupational AI-exposure scores (following felten2023) matched to each metro's occupational composition prior to the sample period. The score is time-invariant by construction, reflecting the occupational structure of each metro rather than any post-treatment adaptation. In the regression analysis, the index is standardised to mean zero and unit standard deviation.

# 2 Data Construction Pipeline

The raw data are processed in three sequential steps, implemented in Python (`parse_and_merge.py` and `build_panel.py`).

1. **Parsing.** The IPUMS fixed-width `.dat` file is parsed using column specifications from the XML codebook. Decimal scaling is applied to variables with implied decimal places. The result is written to `Data/Processed/usa_00004.csv` (≈35 million rows across 11 survey years).
2. **Merging auxiliary data.** The parsed ACS file is merged with the FHFA HPI by `MET2013` × `YEAR`, using the MSAD crosswalk where necessary. The AI exposure index is then merged by `MET2013`. This produces the enriched microdata file `Data/Processed/usa_00004_merged.csv`.
3. **Collapsing to panel.** Individual-level records are aggregated to a metro-year panel using population weights (`PERWT`). Migration flows are identified for metro-to-metro movers (see Section 4). The result is `Data/Processed/metro_year_panel.csv`.

# 3 Sample Restrictions

The following restrictions are applied when building the panel:

- **Non-metropolitan areas excluded**: observations with `MET2013` $= 0$ are dropped.
- **Institutional group quarters excluded**: observations with `GQ` $\in \{3, 4\}$ (correctional facilities and other institutional housing) are dropped; non-institutional group quarters (`GQ` $= 5$, e.g. college dormitories) are retained.
- **Working-age adults**: the sample is restricted to individuals aged 25–55.
- **Balanced panel**: only metropolitan areas observed in every year of the relevant sample are retained, ensuring a balanced panel for the two-way fixed-effects estimator.

Table 1 summarises the three sample specifications used in the analysis.

Table 1: Sample specifications

| Specification | Years included | Metros | Observations |
|---|---|---|---|
| All years | 2014–2024 (11 years) | 242 | 2,662 |
| Drop 2020 | 2014–2019, 2021–2024 (10 years) | 242 | 2,420 |
| Drop 2020 & 2021 (main) | 2014–2019, 2022–2024 (9 years) | 242 | 2,178 |

*Notes*: 2020 is excluded in two specifications because the Census Bureau used experimental weighting procedures in that year. 2021 is additionally excluded in the main specification as a COVID-19 recovery year with atypical migration patterns. In all specifications the pre-treatment period ends in 2022 and the post-treatment period covers 2023–2024.

# 4 Migration Variable Construction

Inter-metropolitan migration is identified using the ACS migration variables. A respondent is classified as an **in-migrant** to metro $m$ in year $t$ if: (i) `MIGRATE1` $\in \{2, 3, 4\}$ (moved within the US in the past year); (ii) their origin metro `MIGMET131` is a valid metropolitan area ($\neq 0$); and (iii) their origin and current metros differ (`MIGMET131` $\neq$ `MET2013`). The same movers are used to compute **out-migration** from their origin metro, identified by grouping on `MIGMET131`. All counts are weighted by `PERWT`.

# 5 Variable Definitions

## 5.1 Outcome Variables

See 2

## 5.2 Control Variables

see 3

## 5.3 AI Exposure

The AI exposure index (`ai_exposure`) is a metro-level, pre-determined measure constructed as the employment-weighted average of occupation-level AI-exposure scores across all occupations present in the metro. The score reflects the occupational composition of each metro prior to the ChatGPT shock and is held fixed across time. Across the 242 metros in the balanced panel, the index ranges from $-0.465$ to 0.232 with a standard deviation of 0.125. In regressions, the variable is standardised (`ai_std`) to mean zero and unit standard deviation so that all coefficients represent the effect of a one-standard-deviation increase in AI exposure.

Table 2: Outcome variable definitions

| Variable | Name in panel | Definition |
|---|---|---|
| In-migration rate | `inmig_rate` | Weighted in-migrants ÷ population |
| In-migration rate (BA+) | `inmig_rate_ba` | Weighted BA+ in-migrants ÷ population |
| In-migration rate (no BA) | `inmig_rate_no_ba` | Weighted non-BA in-migrants ÷ population |
| Skilled in-migrant share | `skilled_inmig_share` | BA+ in-migrants ÷ total in-migrants |
| Net migration BA+ (rate) | `net_mig_ba_rate` | (BA+ in-migrants − BA+ out-migrants) ÷ population |
| Net migration no BA (rate) | `net_mig_no_ba_rate` | (non-BA in-migrants − non-BA out-migrants) ÷ population |
| Incumbent share BA | `incumbent_share_ba` | BA+ among non-movers ÷ non-mover population |
| In-migration rate (young) | `inmig_rate_young` | Weighted in-migrants aged 25–35 ÷ population |
| In-migration rate (old) | `inmig_rate_old` | Weighted in-migrants aged 45–55 ÷ population |

*Notes*: BA+ is defined as `EDUCD` $\geq$ 101 (bachelor's degree or higher). Incumbent share BA uses total BA-weighted population minus in-migrant BA count in the numerator, and total population minus total in-migrants in the denominator.

Table 3: Control variable definitions

| Variable | Name in panel | Definition |
|---|---|---|
| Share with BA+ | `share_ba` | Weighted BA+ residents ÷ population |
| Log population | `log_pop` | Natural log of weighted population (ages 25–55) |
| Mean age | `mean_age` | Weighted mean age |
| Female labour force share | `share_female_lf` | Weighted female labour force participants ÷ labour force |
| Unemployment rate | `unemp_rate` | Weighted unemployed ÷ labour force |